

3D occupancy grid mapping using statistical radar models

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Abstract—We have developed a numerically efficient occupancy grid mapping method in three dimensions for automotive radar, where we take into account the radar measurement signal-to-noise ratio. The mapping performance, i.e. to estimate length, height, and in-between spacing of parked cars, is demonstrated as we use acquired data from a radar prototype developed in collaboration with Qamcom Research and Technology³. The radar has a unique antenna providing unambiguous azimuth and elevation for a wide field of view radar, covering $\pm 50^\circ$ in both dimensions, making mapping in three dimensions feasible. Employing self-developed off-line radar signal processing on raw data, we extract SNR which is used together with a Swerling 1 model to compute the probability of detection for grid map update. Moreover, we present a novel very simplistic way of updating the grid as we use fast trilinear interpolation in the measurement domain, in which the grid spacing is uniform. Having mounted the radar in forward direction the EGO-vehicle drive parallel to four parked cars with different inter-spacing, and we manage to measure the distances within the error of the grid spacing, 0.2 m.

I. INTRODUCTION

As the automatic functions in today's automobile increases in complexity the need for robust sensors is inevitably also becoming more important. This paper will focus on low-cost automotive radar sensors, and their ability to create an image of the environment. Radar sensors are, compared to optical sensors like scanning lasers, very robust to bad weather, dust and particles, vibrations and rough conditions in general. They also possess the ability to measure radial velocity instantly, which facilitates reliable detection of moving objects. In addition they are quite cheap as compared to scanning lasers. Of course, they have some disadvantages, like interference problems and relatively low resolution. In this paper we will target the resolution (and accuracy) issue of the radar, and show the potential of radar 4D measurements (range, doppler, azimuth and elevation) by forming a 3D image of the stationary environment, by using what is called occupancy grid mapping.

Much of the work in today's automotive safety systems is related to autonomous functions, like autopilots for highways or even autonomous drive in urban environment. There exists two main approaches for making the vehicle understand what is the environment in the autonomous drive research, one where sensor data is refined to features representing some part of the environment. Using this information the vehicle can make decision about where to drive and where not to.

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However, mimicking the human driver is extremely difficult due to the high complexity of the environment. The second, and presumably more reliable approach [4], is to rely on detailed maps which combined with sensor readings localizes the vehicle, and we obtain very useful prior information of the environment. This has been researched for decades, starting with indoor robotics, and is usually called SLAM (simultaneous localization and mapping). Our work aims to investigate the possibility of the commercial low-cost automotive radar sensors playing a bigger role in the localization and mapping systems. The contribution is making occupancy grid in 3D using radar, which is much thanks our ability to measure both azimuth and elevation accurately. Also, we have a novel numerically efficient procedure for handling the inverse sensor model in the grid map update using trilinear interpolation, which enables for real-time processing of 3D data.

II. BACKGROUND

As mentioned, localization and mapping is a well studied field of research and there exists vast number of publications starting at the previous century. Radar-SLAM using extended Kalman filter for interleaved mapping and EGO-vehicle localization was performed by Clark in 1999 [3]. Here, the environment is represented by a discrete set of landmarks which is the traditional representation. The association between measurements and landmarks needs be correct all the time or else the vehicle pose will get a bias error (and the map will get distorted). To handle the association problem there exists multi-hypothesis approaches such as FastSLAM [11] and [15]. However, for radar SLAM it is common to circumvent the problem by representing the environment as a grid and apply scan matching. A nice approach was taken by [16] where they use a FMCW radar and signal power down range (no detections) for each measured bearing rather than the set of detected plots. Scan matching works in 2D and is convenient when the sensor is mechanically scanning a narrow lobe, but it requires a high rotation frequency (or low vehicle speed). Actually, some years earlier [5] performed 3D radar mapping also with radar, but with a pencil beam mechanically scanned in both azimuth and elevation (similar to a Lidar).

More recent work from [17] show the potential of radar scan matching as a great way of circumventing the association problem which arises in the landmark-based EKF approaches. They use a 24 GHz 360° 2D scanning FMCW radar. However, the scanning rate is 1Hz which is definitely a showstopper when it comes to automotive. Localization is made by cross correlation and Fourier-based

image registration technique [6]. The correlation is in three dimensions since we have both azimuth deviation, and lateral plus longitudinal displacement. The iterative-closest point is another way to do this, as showed in the indoor radar SLAM from TNO [10] using 24 GHz radar to assist fire-fighters encountering vision-denied environments. Another way is to use to normal distributions transform [2] which was used in automotive by [14].

Despite the fact that we have argued that the radar sensor, in contrast to Lidar, has the advantage of efficiently covering signal strength of all reflective objects down range, we will not use that here. Instead we will use a discrete set of detections to form the scans used for matching. The reason is primarily that automotive radar sensors are not mechanically scanning a narrow lobe. To assure a high update rate (which is required in automotive) we use a wide lobe, detect objects in Doppler-range with CFAR adaptive detectors [18], and compute the angles for that cell using DOA techniques. We could have assumed all environment to be stationary and used the fact that each Doppler channel represent an angle in azimuth (this would only work for 2D). However, this would only work for side-looking radar and since we are looking forward, objects at a particular angle to the right merge with objects at the same angle to the left. Thus, we form a measurement scan using detected plots, and update the grid map with our inverse sensor model, which is described in section IV. The scope of this paper is limited to the mapping problem, and no localization will be performed. Using real data we show promising 3D mapping results with only one single forward-looking radar prototype.

The motivation for mapping the environment in 3D is primarily to support localization better, especially in hilly driving situations. Obviously, it is also beneficial when entering tunnels and passing road signs located above the road. When the stationary environment consist of merely guard rails and light poles, it can be difficult to localize the vehicle in the longitudinal (driving) direction if there are no curves. When using 3D mapping, the altitude information will discriminate the light poles, and facilitate robust longitudinal localization.

III. RADAR PROTOTYPE AND SIGNAL PROCESSING

The field test were made using a 77 GHz prototype radar from Qamcom R&T, see figure 1. This radar is a 77 GHz radar providing digital scanning in both azimuth and elevation. The size of the receiving aperture is along the horizontal and vertical cut about 8λ and about 2λ , respectively. Using a simple rule of thumb, derived from the Cramr Rao analysis, DOA estimation accuracy can be written as [1]

$$DOA_{acc} = \frac{\lambda}{\sqrt{2D}\sin(\theta)SNR} \quad (1)$$

where λ is the wavelength, D is the maximum distance between the phase centers, θ is the scanning angle and SNR is the signal to noise ratio. Using an SNR of 20 dB

and a scanning angle of 20° , the DOA accuracy becomes $\approx 0.3^\circ$. The resolution can either be computed from the Fourier resolution or using the Cramr Rao lower bound (CRLB). The Fourier resolution can be seen as the upper bound or worst case. Here, we observe that the lower limit (CRLB) of the vertical resolution performance is about 3° . The corresponding result along the horizontal cut is about 0.5° . The advantages and uniqueness of this Qamcom radar is, although this very wide field of view, the provided unambiguous angular estimation of the targets in both elevation and azimuth in $\pm 50^\circ$. Furthermore, the system bandwidth of the radar system is about 1.5 GHz (75.7-77.2 GHz) where a flexible waveform selection is possible to employ. Finally, the prototype system also provides raw data acquisition at a duty factor of 40 %. This high bite rate acquisition (5 Gbit/s) offers an unique possibility to analyze complex scenarios.



Fig. 1. Qamcom prototype radar mounted on our test vehicle.

The signal processing employed for the target detection is based on the conventional CFAR adaptive detection framework working on a single channel [20]. That is, we employ a certain number of training data to adaptively compute the threshold of the cell under test, see figure 2. To obtain this single channel we extract the range and Doppler outcomes by computing FFT:s along both fast time, i.e., along range frequency samples, and slow time, i.e., along the chirps. Thereafter we sum the range-Doppler results obtained for each receiver channel incoherently, i.e., sum by power. Moreover, in figure 2 we also show a typical range-Doppler outcome for the very wide field of view case. Note that, to obtain CFAR behavior we have employed a training data set that covers both range and Doppler cells. Thus, we employ guard bins in both the range and Doppler dimensions.

Next, in range and Doppler the target estimation is merely given by the interpolated outcome of the achieved detection. The spatial estimation in azimuth (or directional cosine u) and elevation (or directional cosine v) is based on the MUSIC method, i.e., a high resolution method.

The computation of detection probability used for the occupancy grid herein is relying on a generic form of the radar equation combined with the Swerling 1 case model of the target RCS fluctuations is used. The Swerling 1 case model of the target is as follows: The probability of detection is computed by

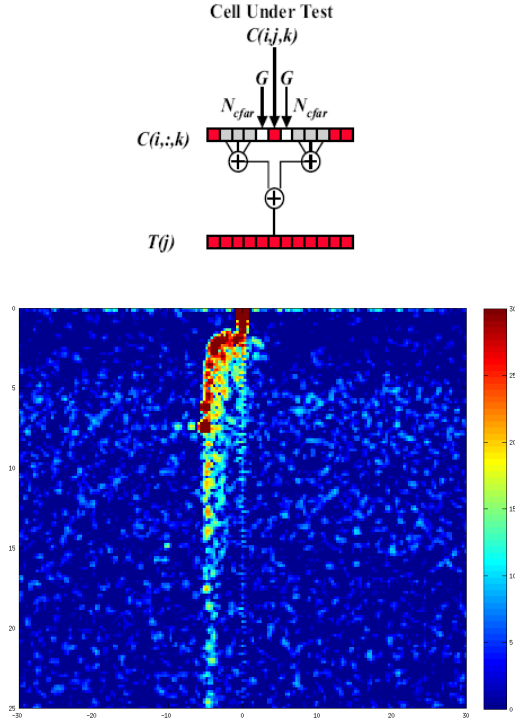


Fig. 2. The top figure illustrates a CFAR detector based on a single channel. The N_{CFAR} is the training data set and G denotes the guard bins. Image courtesy of Lincoln Laboratory, MIT. The bottom insert shows a typical range Doppler outcome using very wide field of view. On the x-axis we have Doppler channels and range is displayed from top to bottom.

$$P_d^{\text{Swerling}}(SNR) = P_{fa}^{\frac{1}{1+SNR}} \quad (2)$$

$$SNR = SNR_{50} \frac{\sigma}{\sigma_{50}} \left(\frac{R_{50}}{R} \right)^4 \quad (3)$$

where the constants σ_{50} , SNR_{50} and R_{50} are normalized to the probability of detection equal to 50%, and $P_{fa} = 10^{-3}$. The radar cross section of the target σ can be computed from the estimated SNR . In this paper we will only use equation (2).

IV. OCCUPANCY GRID MAPPING BY TRILINEAR INTERPOLATION

There are basically two ways to representing the world, with a mixture of states, or with a discrete grid. When tracking multiple moving objects it is convenient to assign a state to each object. When mapping stationary surroundings both representations can be used. Some benefits can be achieved by interleaving the estimation of both moving and stationary objects [8]. However, since we have a Doppler radar we can classify measurements as moving or stationary, and for simplicity we assume separate estimators. In this paper we will focus on mapping the stationary environment. By choosing occupancy grid maps [13] we avoid solving the association problem, i.e. assigning measurements to states (and the merging problem in PHD-filter [15]). Moreover, we

make no assumption about the geometry. Since the resolution of the radar is quite high, a Gaussian distribution may not be an appropriate model of stationary objects in urban environment.

The main drawback of occupancy grid map is the high memory consumption, and the difficulty in estimating the occupancy state of the grid cell. In general, occupancy grid mapping recursively compute the *probability* of each grid cell being occupied, rather than *estimating* the binary state. However, there are approaches for estimating the state of each grid map using expectation maximization (EM) [19]. For our three-dimensional grid it would be quite computationally-heavy, thus we restrict ourselves to computing the probability. We denote the binary stochastic grid variable by $\mathbf{m}_i$ on the grid domain $\mathcal{G}$ with an estimated probability of being occupied

$$P(\mathbf{m}_i) : \mathcal{G} \rightarrow \mathbb{R}[0, 1] \quad (4)$$

where the grid can be both in measurement domain $\mathcal{G}_{\text{measurement}}$ and in the cartesian domain $\mathcal{G}_{\text{cartesian}}$. The two sets are defined by

$$\mathcal{G}_{\text{cartesian}} = \{x_c, y_c, z_c\} \quad \Delta x = \Delta y = \Delta z = 0.2m \quad (5)$$

$$\mathcal{G}_{\text{measurement}} = \{r_m, \theta_m, \phi_m\} \quad (6)$$

$$\Delta r = 0.3m \quad 0 < r_p < 30m$$

$$\Delta \theta = 1.2^\circ \quad -^\circ < \theta_m < 60^\circ$$

$$\Delta \phi = 2^\circ \quad -20^\circ < \phi_m < 20^\circ$$

where Δ indicate grid spacing.

The resolution of the measurement grid $\mathcal{G}_{\text{measurement}}$ is balanced to resolution of the measurement, see section III. Assume that we know the EGO-motion perfectly, and thus know the sensor location $[x_s, y_s, z_s, h_s]$ (neglecting pitch). Then we can use the standard approach which breaks the problem of making the map into sub-problems by assuming independence between cells. This binary Bayes filter estimates the occupancy probability for each cell very efficiently. It is Bayes in the sense that both measurements and grid map occupancies are stochastic, but the vehicle pose is deterministic. However, there are approaches that take into account the uncertainty in the vehicle pose using particles [9]. Rather than filtering the occupancy map posterior $P(\mathbf{m}_i | \mathbf{Z}_t)$ probability, we form the ratio and take the logarithm. We will skip the details here, see [12] for derivation, and just write down the update equation using log-odds ratio L_t

$$\underbrace{\log \left(\frac{P(\mathbf{m}_i | \mathbf{Z}_t)}{1 - P(\mathbf{m}_i | \mathbf{Z}_t)} \right)}_{L_t} = \underbrace{\log \left(\frac{P(\mathbf{m}_i | \mathbf{Z}_{t-1})}{1 - P(\mathbf{m}_i | \mathbf{Z}_{t-1})} \right)}_{L_{t-1}} + \underbrace{\log \left(\frac{P(\mathbf{m}_i | \mathbf{Z}_t)}{1 - P(\mathbf{m}_i | \mathbf{Z}_t)} \right)}_{\text{new information}} - \underbrace{\log \left(\frac{P(\mathbf{m}_i)}{1 - P(\mathbf{m}_i)} \right)}_{\text{prior}=0} \quad (7)$$

for the map $\mathbf{m}_i$ and all the detections $\mathbf{Z}_t = \{Z_1, Z_2, \dots, Z_t\}$. If we have a prior of the map it is very easy to incorporate

in (7). Here, we will assume no prior information. Then all we need for the update is $P(\mathbf{m}_i|\mathbf{z}_t)$ which is called the *inverse sensor model*. Unlike most implementations, we will use the measurement signal-to-noise ratio (SNR) instead of fix probability of detection and convert SNR to P_d using (2). We will form the (inverse) sensor model using a discrete set of detections $Z_t = \{z_1, z_2, \dots, z_N\}$ whose SNR is above $SNR_{50} \approx 9, 6$, which is the SNR corresponding to $P_d = 0.5$. This is in addition to the adaptive CFAR detector described in III. The SNR map is converted to P_d as the log-odds ratio is updated by the “new information”

$$L_t = L_{t-1} + \log \left(\frac{P_d^{\text{Swerling}}(M_t)}{1 - P_d^{\text{Swerling}}(M_t)} \right) \quad (8)$$

where the variable M_t is a grid map of signal-to-noise ratio.

$$M_t : \mathcal{G} \rightarrow \mathbb{R} [-\infty, \infty] \quad (9)$$

Like many implementations we will have some assumptions of free space in front of the sensor, and elsewhere (unknown) we will have the SNR_{50} . The free space assumption more or less heuristic and is written as

$$M_t^0(r_m, \theta_m, \phi_m) = SNR_{50} - (SNR_{50} - SNR_{\text{free}}) \cdot \exp \left(-\frac{1}{2} \frac{r_m^2}{r_0^2} - \frac{1}{2} \frac{\theta_m^2}{\theta_0^2} \right) \quad \forall r_m, \theta_m, \phi_m \in \mathcal{G}_{\text{measurement}} \quad (10)$$

where SNR_{free} corresponds to probability of free grid cell $P_{\text{free}} = 0.45$, and the extension of the free space is parametrized by $r_0 = 6m$ and $\theta_0 = 30^\circ$. The free space assumption is identical for all elevation layers $-20^\circ < \phi_i < 20^\circ$.

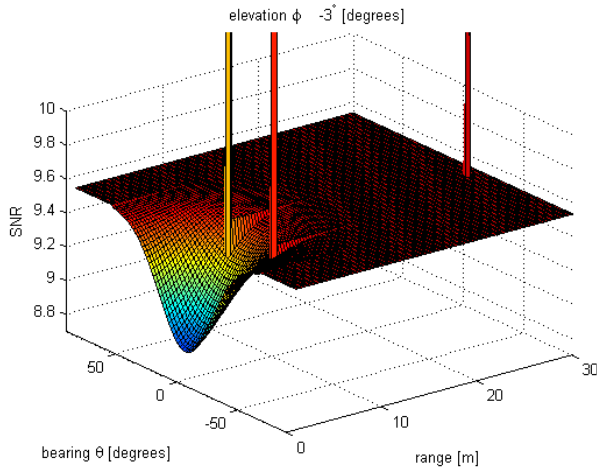


Fig. 3. The measurement grid $M_t(r_m, \theta_m, \phi_m)$ for a certain elevation layer at $\phi_m = -3^\circ$. The image is zoomed to show the empty search prior SNR_{free} and SNR_{50} . The two near detections have SNR of 55 (bore-sight) and 100, respectively. The detection at 30m has a SNR of 13.

The measurement M_t is formed by truncating measurements Z_t into the nearest grid point in $\mathcal{G}_{\text{measurement}}$ and add their SNR (if higher SNR_{50}). In Figure 3 we show an example of one elevation layer of the measurement M_t

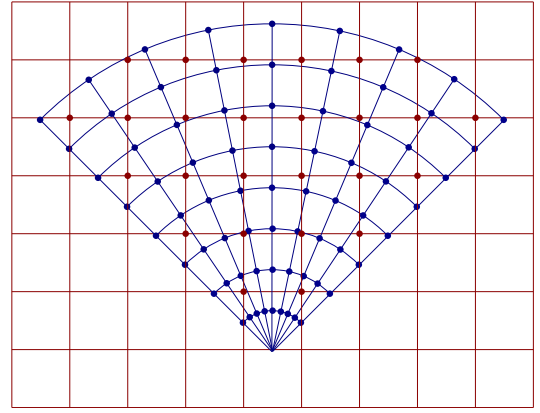


Fig. 4. Visualization of the Cartesian (red) grid $\mathcal{G}_{\text{ROI}}$ and polar (blue) $\mathcal{G}_{\text{measurement}}$ grids used in interpolation in the 2D case. The red dots constitute the subset of $\mathcal{G}_{\text{ROI}}$ that is interpolated using $\mathcal{G}_{\text{measurement}}$ and their values $M_t(r_m, \theta_m, \phi_m)$. The rest of $\mathcal{G}_{\text{ROI}}$ are assigned to SNR_{50} .

model for $\phi = 3^\circ$. Measurement noise is usually added to the measurement grid M_t by convolution, but in our case we have skipped this step, and instead use a resolution with grid size in the same order as the theoretical measurement noise. This to reduce computational load.

The numerically complex part of occupancy grid mapping is to map the (spherical) measurement grid $\mathcal{G}_{\text{measurement}}$ to the Cartesian grid $\mathcal{G}_{\text{cartesian}}$. Each Cartesian cell to be updated detection needs the value of the most overlapping measurement cell. Although GPUs can handle this fairly well [7] we believe that an interpolation scheme is far more simple and better, especially when it comes to numerical performance. Interpolation is only fast for a equally spaced monotonically increasing grid. This means that *the Cartesian grid must be converted to the measurement grid*. The measurement is not uniform in the Cartesian space. Hence, interpolation is made by selecting a region of interest around the vehicle from the Cartesian grid points $\mathcal{G}_{\text{ROI}} \subset \mathcal{G}_{\text{cartesian}}$ and compute their location in the spherical grid $\{x_c, y_c, z_c\} \iff \{r_c, \theta_c, \phi_c\}$ by subtracting the sensor location $[x_s, y_s, z_s, h_s]_t$ and converting to spherical coordinates by

$$r_c = \sqrt{(x_c - x_m)^2 + (y_c - y_m)^2 + (z_c - z_m)^2} \quad (11)$$

$$\theta_c = \tan^{-1} \left(\frac{y_c - y_m}{x_c - x_m} \right) - h_s \quad (12)$$

$$\phi_c = \tan^{-1} \left(\frac{z_c - z_m}{\sqrt{(x_c - x_m)^2 + (y_c - y_m)^2}} \right) \quad (13)$$

where h_s is the (known) heading of the sensor. The update is then performed by interpolating the measurement grid to the Cartesian grid

$$M_t(x_c, y_c, z_c) = M_t(r_c, \theta_c, \phi_c) = I(\{r_m, \theta_m, \phi_m, M_t(r_m, \theta_m, \phi_m)\} \rightarrow \{r_c, \theta_c, \phi_c\}) \quad \forall r_c, \theta_c, \phi_c \in \mathcal{G}_{\text{ROI}} \quad (14)$$

by using trilinear interpolation (two bilinear interpolations combined with a linear interpolation). All points in $\mathcal{G}_{ROI}$ not covered by the current measurement grid $\{r_m, \theta_m, \phi_m\}$ get the value SNR_{50} which corresponds to $P_d = 0.5$. In the two-dimensional case this is visualized in Figure 4 where the values at the red dots are computed using the nearest 4 blue dots using bi-linear interpolation. This means that there may be several detections near the radar that does not update the map. However, most of the detections at short range are due to DC-noise. On the other hand, our approach will avoid holes in the Cartesian grid at larger distances, where the measurement grid cells are significantly larger. Hence, there are other advantages with this approach than purely the numerical.

Regarding the numerical part, it is obvious that replacing the angles azimuth and elevation by the directional cosines, will make some improvement. In fact, most radar SP will compute the cosines rather than angles since it is more intuitive for a phased array as the measurement noise is constant and not depending on angle.

V. RESULTS

We performed the measurements at NEVS (former Saab) test track in Trollhättan. In the test scenario the EGO-vehicle was moving at constant velocity at about 20 km/h where the parked cars, were positioned 2 meter to the right of the EGO-car. The parked cars were a Lexus GS, a Volvo S60, a Toyota Prius, and a Volvo V60, see Figure 5. The sensor was mounted on a Volvo S60 Thule Conduit box positioned 1.5 meters above the ground in the vehicle front, see Figure 6. We also have a reference position D-GPS system with gyro mounted in the back seat. Measurements were made using a chirp bandwidth of 1 GHz (76-77GHz), accumulating 512 pulses of $40\mu s$, with a duty of 40%. This corresponds to a update rate of 20 Hz in theory, which in reality was 17 Hz. At such low speeds it is more than enough.



Fig. 5. The four parked cars as seen by the camera mounted on the radar.

Deriving quantitative results for this type of study is not straight-forward. We have chosen make distance measurements in the grid map manually, which has a variability of approximately 0.1 m depending on who is measuring. But

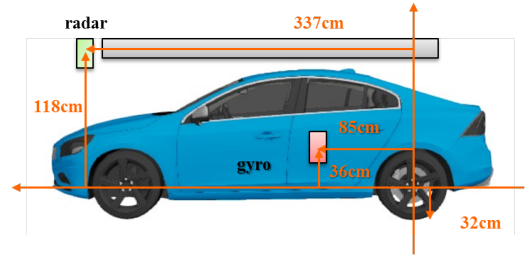


Fig. 6. The mount of the radar prototype on the S60 test vehicle.

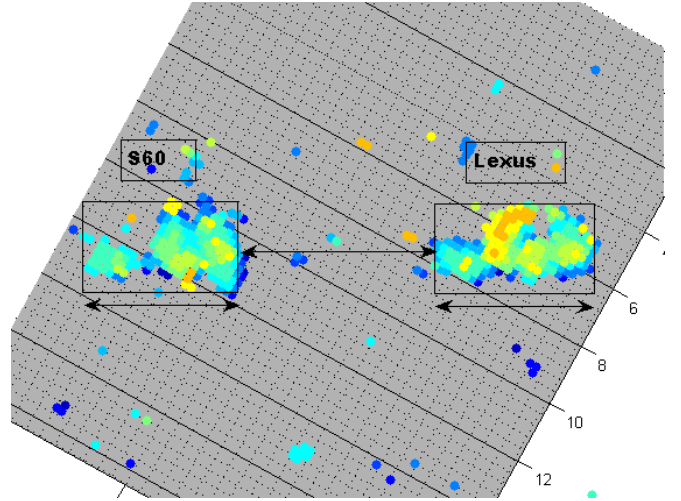


Fig. 7. Top down view of the 3D grid of the first two vehicles. The dots represent cells having a log-odds ratio $L_t > 7$ and their color indicate height. Boxes are manually placed and form basis for distance measurements.

first we must decide for a log-odds ratio (or corresponding P_d) threshold at which we have confirmed an occupied cell. If we chose all cells having $L_t > 1$ which corresponds to $P_d = 0.5$ (and SNR_{50}) we will be sensitive to noise and may also over-estimate the size of objects. We set the level for the log-odds ratio to $L_{confirmed} = 7$, which may seem high since it translates to $P(\mathbf{m}_i | \mathbf{Z}_i) > 0.999$ but it actually corresponds to three detections with $SNR = 18$ ($P_d = 0.9$) or six detections with $SNR = 13.6$ ($P_d = 0.75$) so this threshold is quite adequate. Using this threshold we find occupied cells and use a top-down view of the cars to manually measure vehicle size and inter-vehicle spacing, see Figure 7. The result is presented in Table I. Apart from the length of the Prius we have an error of 5-10 cm. Adding the variability in the manual estimate, we conclude that the error in average is of the same size as the grid spacing, 20 cm.

We estimated the numerical load in Matlab which gives a rough estimate, given that we are on a PC Intel core i7 platform. On a radar signal processing platform we would use a different microprocessor, dedicated for interpolation. Nevertheless, we see that the interpolation for a $100 \times 100 \times 20$ grid $\mathcal{G}_{measurement}$ takes about 0.1 seconds, where most of the time is spent on basic arithmetics and checking out-of-regions. We use our own implementation which is about 3 times faster than Matlab's. The inverse tangent function for

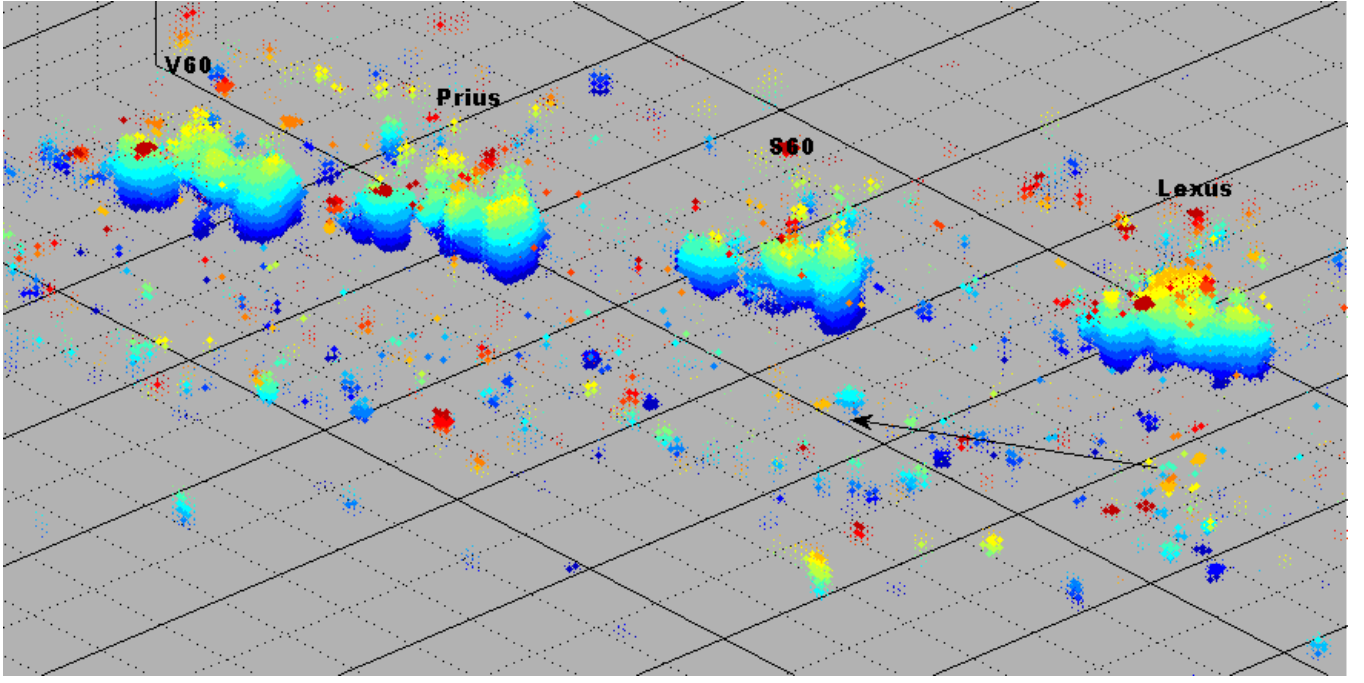


Fig. 8. A 3D view of the parked cars and the spacings where each voxel (cartesian 3D grid element) with likelihood ratio larger than zero is visualized with a dot. The larger the ratios, the larger the size of the dot. Thus higher occupancy probability becomes more prominent. Blue color represents the lower levels and red color higher levels.

item	true length	est. length	true height	est. height
car 1 (Lexus)	4.85 m	4.7 m	1.45 m	1.6 m
spacing 1	6.00 m	5.9 m		
car 2 (S60)	4.63 m	4.7 m	1.48 m	1.4 m
spacing 2	4.00 m	4.0 m		
car 3 (Prius)	4.45m	4.7 m	1.49 m	1.4 m
spacing 3	2.00 m	1.9 m		
car 4 (V60)	4.64 m	4.6 m	1.48 m	1.4 m

TABLE I

ALL THE DISTANCES THAT WAS MEASURED IN THE GRID MAP.

the $\mathcal{G}_{ROI}$ grid takes much less time, 0.01 s. Worth noticing is that the power functions in the Swerling model takes 0.05 s, and the logarithm in the update take 0.025 s. An easy fix for the Swerling calculation is to convert the SNR of the measurements directly when making M_t .

In addition to the results in the table we present a 3D view of the scene in Figure 8, using variable sized point cloud. The larger ratio, the bigger sized point. The color of the points indicate height, and due to ground reflections we will get some occupied cells below ground. In fact, the two bottom layers (dark blue) of the vehicles are at -0.4 m and -0.2 m.

VI. DISCUSSION

Despite the fact that the sensor used in this study is the first prototype radar from Qamcom R&T, it is fully functional and the DOA estimation works satisfactory which is quite extraordinary. The only problem is that the transmit lobe is slightly too wide, $\pm 70^\circ$ in both azimuth and elevation, since the receiving antenna design supports $\pm 50^\circ$ unambiguous estimation. This was due to a manufacturing error, and it was

fixed to the second prototype which has a good and flexible transmit antenna pattern. The noise, especially on the left side of the road at higher elevations in figure 8, are most likely due to the fact that we transmit outside unambiguous region. Decreasing the field of view to the unambiguous interval and mount the radar in the side-looking direction is one of our primary next steps. In the range-Doppler map in figure 2 each Doppler channel contains a mix of stationary objects to the left and right of the road having the same radial velocity. By looking sideways, we avoid this problem and are able use the Doppler (together with EGO speed) for a rough DOA estimate.

Finally, we would like to comment on the way the measurement grid M_t is constructed using binning of the measurement Z_t . We could have used tri-linear interpolation, similarly as for the grid update, which would make the signal strength to be spread over several grid cells in M_t . Or, like the most commonly used approaches, increase the measurement grid resolution significantly and use the measurement noise as a kernel to perform convolution of the grid. Sure, this makes sense, but our point with this paper is that we would rather use the estimated SNR to control the impact of each measurement to the grid, than smear the data in the spatial domain based on measurement uncertainty. Having an adaptive P_d we will reduce the effect of plots for which the direction estimation is uncertain since their SNR is usually significantly lower. When the detector threshold is low, the measurement noise in the angular domain is heavy-tailed, meaning that the estimated angle may be completely wrong. Then, adaptive P_d is more useful than smearing over a large angular range.

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